HazardGuard C3: Multi-Hazard Emergency and Safety Automation

Gas leak, fire, danger-zone and tank-overflow protection with automatic exhaust / gas cut-off and a phone dashboard

Category	Level	Main board
Electronics	High school	ESP32-C3

1. ABSTRACT

HazardGuard C3 is a low-cost safety controller for kitchens, hotel boiler rooms and workshops. An MQ gas sensor detects LPG and smoke, a DHT11 measures temperature for fire and overheating, and an ultrasonic sensor watches a danger zone around a machine or the level in a tank. When something is dangerous, the ESP32-C3 switches on a relay (exhaust fan or gas shut-off valve), flashes a red LED, sounds a buzzer and shows a warning code on a 4-digit display. The controller also creates its own WiFi hotspot, so any phone can open a live dashboard at 192.168.4.1 to watch the readings, switch between Household, Hotel and Industrial profiles, change thresholds and silence the alarm.

2. AIM OF THE PROJECT

- Detect gas leaks, smoke, fire / overheating, danger-zone breaches and tank overflow with one controller.
- Act automatically: run an exhaust fan or close a gas valve through a relay.
- Warn people with LEDs, buzzer and a display, and on any phone without internet.
- Allow different safety limits for homes, hotels and factories, saved even after power cuts.

3. PROBLEM STATEMENT AND NEED

LPG leaks and kitchen fires cause many accidents every year, often because nobody notices in time. Commercial gas alarms only beep - they do not ventilate the room or cut the gas - and they cannot be checked from another room. HazardGuard C3 reacts on its own within a fraction of a second and shows the situation on any phone nearby.

4. COMPONENTS USED AND COST

Component	Model	Qty	Purpose	Approx. cost
Controller	ESP32-C3 SuperMini	1	Reads sensors, runs logic, WiFi hotspot	Rs. 300
Gas sensor	MQ-2 / MQ-135 module	1	LPG, methane and smoke	Rs. 160
Temperature sensor	DHT11	1	Fire / overheating, humidity	Rs. 110
Ultrasonic sensor	HC-SR04	1	Danger-zone breach and tank level	Rs. 90
Relay	5 V 1-channel, opto-isolated	1	Switches exhaust fan or gas solenoid	Rs. 70
Display	TM1637 4-digit	1	Readings and warning codes	Rs. 90
Buzzer	5 V active buzzer	1	Audible alarm	Rs. 20
LEDs	Green, yellow, red 5 mm	3	Safe / warning / danger	Rs. 10
Resistors	3 x 220 Ω, 1 kΩ, 2 kΩ, 10 kΩ	1 set	LEDs, echo divider, DHT pull-up	Rs. 15
Power	5 V 2 A USB-C adapter	1	Supply	Rs. 150
Total				Rs. 1,015

Table 1: Bill of materials (approximate local prices)

5. CIRCUIT CONNECTIONS

Part pin	ESP32-C3 pin	Notes
MQ gas AO	GPIO 1 (ADC)	Through a 10k:10k divider if the module runs on 5 V
TM1637 CLK / DIO	GPIO 2 / 3	
HC-SR04 TRIG / ECHO	GPIO 4 / 5	ECHO through 1 kΩ / 2 kΩ divider
DHT11 DATA	GPIO 6	10 kΩ pull-up to 3.3 V
Buzzer	GPIO 7	
Relay IN	GPIO 10	Active LOW module
Green / yellow LED	GPIO 0 / 8	220 Ω to GND
Red LED	GPIO 9	3.3 V -> 220 Ω -> LED -> GPIO 9 (active LOW, safe for boot)

Table 2: Wiring

6. SYSTEM OVERVIEW

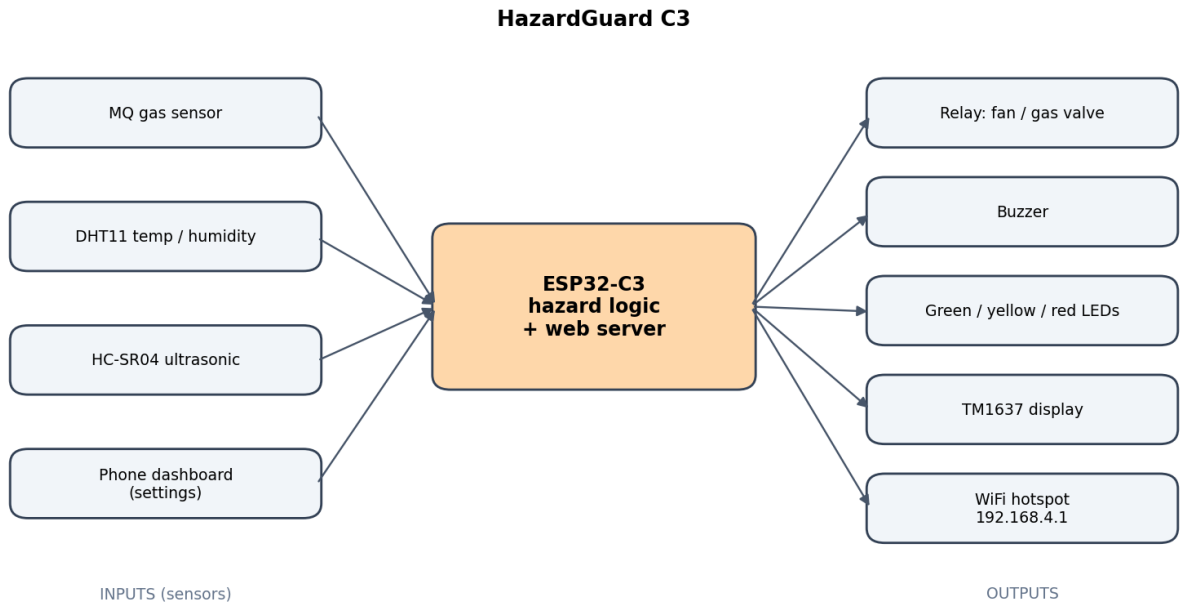


Figure 1: Block diagram

7. HOW I DID THIS (METHOD)

The gas reading is averaged over 8 samples to remove noise. For the first 60 seconds the MQ heater warms up, so gas alarms are held back to avoid false alarms. Each profile has its own limits; the Industrial profile is the most sensitive. If a sensor stops answering, the dashboard shows 'sensor fault' and the display shows E1 or E2 instead of showing a made-up number.

When danger is detected, the relay switches on and stays on for 8 seconds after the danger clears, so a fan does not chatter on and off around the limit. The relay can also be forced on or off from the phone. Thresholds and the chosen profile are stored in the ESP32's flash memory (Preferences), so they survive power cuts.

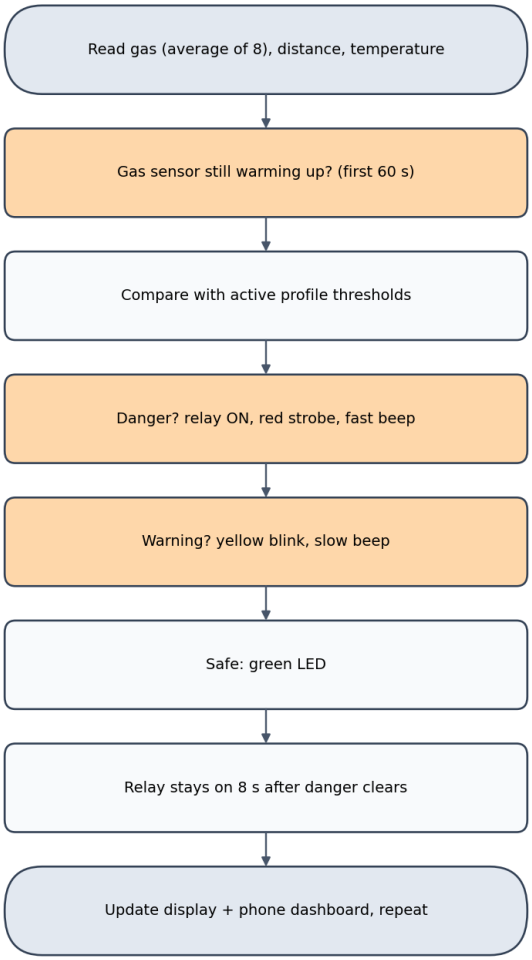


Figure 2: Control flow

8. TESTING AND RESULTS

Figure 3 replays the firmware's decision logic on a simulated LPG leak: the gas reading rises, crosses the warning and danger limits, and falls after the window is opened. The relay switches on at the danger limit and switches off 8 s after the reading drops back, exactly as designed.

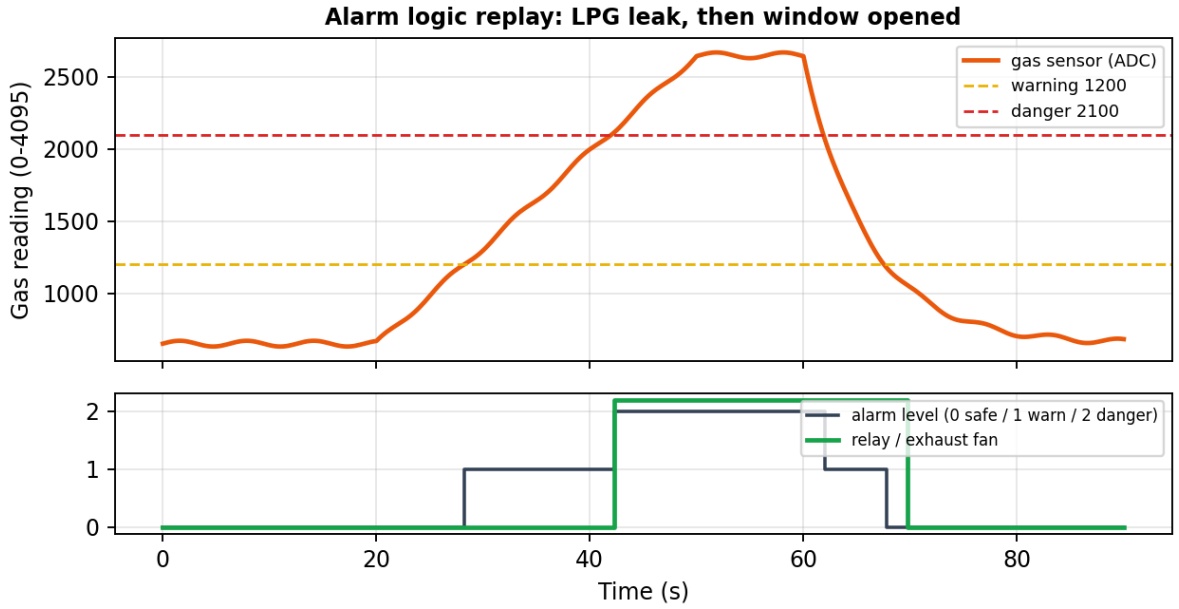


Figure 3: Alarm logic replay

Test on the bench	Expected result
Gas lighter (unlit) near MQ sensor	Yellow, then red; relay ON; 'GAS' flashing
Hot air near DHT11 above 52 °C	Red; 'FIrE' on display; relay ON
Hand within 30 cm of ultrasonic	Danger-zone breach; 'ZonE' on display
Unplug the DHT11	Dashboard shows 'Sensor fault'; display shows E1
Tap 'Silence buzzer' on phone	Buzzer quiet for 2 minutes, lights keep flashing

Table 3: Bench test checklist

9. APPLICATIONS

- Home kitchens: LPG leak alarm with automatic exhaust fan.
- Hotel kitchens and boiler rooms: overheating and gas protection.
- Workshops: keeping people out of a machine's danger zone.
- Tanks and sumps: overflow warning.

10. ADVANTAGES AND LIMITATIONS

- Acts automatically instead of only beeping.
- Works without internet: the phone connects straight to the device.
- Settings change from a phone and survive power cuts.
- Limitation: MQ sensors need warm-up and show relative values, not exact ppm.
- Limitation: the DHT11 reads the air, not flames directly; a flame sensor would react faster.

11. FUTURE IMPROVEMENTS

- Add a flame sensor and a CO sensor.
- Send SMS or app alerts through a home router.
- Battery backup so it keeps working during power cuts.

12. CONCLUSION

HazardGuard C3 combines four kinds of hazard detection, automatic action and a phone dashboard in a controller that costs about a thousand rupees, showing that real safety automation can be cheap and simple.

13. REFERENCES

- Hanwei Electronics - MQ-2 gas sensor datasheet.
- Aosong - DHT11 datasheet.
- Espressif - ESP32-C3 technical reference and Arduino core documentation.